Inheritance of perfect splits in actual wings

Human Compression project: a component of the L019 recovery

Let $G = \langle \rho_0, \ldots, \rho_{r-1} \rangle$ be transitive, with string involutions and every maximal omission intransitive. Suppose it has a perfect split at i, with left and right orbits A, B, and write

$$q = \rho_i = (a\ b)\alpha\beta, \qquad a \in A, \quad b \in B.$$

The marks before i are supported and transitive on A, and those after it are supported and transitive on B. The internal factors α, β fix a, b respectively. A one-point wing has its empty transitive marking.

Lemma 1. *Every maximal omission of either actual wing is intransitive on that wing's support.*

Proof. If a wing remained transitive after deletion of one of its marks, the other full wing and the crossing mark q would join its support to the other support. The corresponding maximal omission of G would then be transitive, a contradiction. □

Lemma 2 (A nonadjacent wing split)**.** *Suppose the right wing has a perfect split at $j \geq i + 2$. Then j is a perfect split of the parent marking, even if $j = r - 1$. Either trivial internal factor remains trivial. The reversed statement holds for a left-wing split at $j \leq i - 2$, including $j = 0$.*

Proof. Write the wing's two sides as C, D, where the marks strictly between i and j are transitive and supported on C, and those after j are transitive and supported on D. Their unique crossing edge is $\{c, d\}$, with $c \in C$, $d \in D$. Every mark after $i + 1$ fixes b by the original unique crossing. The mark ρ_{i+1} moves b, since the right wing is transitive. As it lies before j, its support is contained in C, so $b \in C$.

The parent omission G_j contains groups transitive on A, C, D separately, and q joins A to C through $\{a, b\}$. If q had any transposition from C to D, this omission would be transitive on the full set. Thus q preserves D and its complement setwise. It commutes with ρ_j, and $\{c, d\}$ is the only ρ_j-edge between these two sets. Hence q preserves this edge and fixes each of its endpoints. It also centralizes the transitive group on D generated by the marks after j, so fixes D pointwise. When D is a singleton, preservation already gives the same conclusion; no nontrivial-tail hypothesis is needed.

The marks before j therefore generate a transitive group on $A \cup C$ and fix D, while the marks after j remain transitive and supported on D. The two actual omission orbits are $A \cup C, D$. The j-mark fixes A, so its internal factors are unchanged apart from adjoining fixed points to the first side. This proves both conclusions. □

Lemma 3 (The adjacent split of a sufficiently large wing)**.** *Let the right wing have degree $m = |B|$ and rank $t = r - i - 1$. Suppose $m \geq 2$, $2t > m - 1$, and its first index $j = i + 1$ is a perfect split. Then j is a perfect split of the parent marking and its internal factor on the new left side is trivial. In addition, the original factor β is trivial.*

Proof. The first wing split has a singleton near side and a far side D on which $T = G_{>j}$ is transitive. Its singleton is b. Indeed T fixes b, while only the first wing mark can move it. If $|D| > 1$, the singleton is the only T-fixed point in B; if $|D| = 1$, choose the side names so that the moved point b is the near side. Thus $B = \{b\} \sqcup D$ and $\rho_j = (b\ d)\delta$, where δ is supported on $D \setminus \{d\}$.

The factor β is supported on D, centralizes T, and, if nonidentity, is fixed-point-free there. Its pairs would form a T-invariant system of $|D|/2$ blocks. Choose a subfamily M of the specified T marks transitive on this pair set, with at most $|D|/2-1$ members, using the acquired minimal-transitivity forest. On D, the group $\langle M, q\rangle$ contains the pair flips supplied by $q|_D = \beta$ and is transitive: it is transitive on the pairs and each pair can be interchanged. Add all left-wing marks and ρ_j. The resulting group is transitive on A, joins A to b through q, and joins b to D through ρ_j, hence is transitive on the whole domain. Any T mark absent from M would contradict the parent omission hypothesis. Therefore

$$t - 1 \le |M| \le |D|/2 - 1, \qquad \text{so} \qquad 2t \le |D| = m - 1,$$

contrary to the hypothesis. Thus $\beta = 1$.

The marks through i now act transitively on $A \cup \{b\}$ and fix D, while T is transitive on D and fixes the new left side. The mark ρ_j has only its crossing $\{b, d\}$ on the new left side and fixes all of A. It is therefore a perfect split with trivial left internal factor. This includes a singleton D, where $\beta = 1$ was automatic. Reversal gives the corresponding left-wing assertion. □

Attribution and remaining scope. These are the actual support mechanisms behind the inheritance steps in Cameron–Fernandes–Leemans, *The number of string C-groups of high rank*, arXiv:2212.12723v2, Section 8. The nonadjacent proof makes explicit the parent omission hypothesis and the terminal singleton case. The adjacent proof replaces a classification bound by minimal transitivity on actual involution pairs; it assumes no quotient IP. It also strengthens the usable interface to the stated wing rank condition. Neither lemma claims that an available wing split is internal, or that the full stability correspondence has been recovered.